LAB MANUAL

KARPAGAMCOLLEGE OF ENGINEERING

COIMBATORE – 641032

ANNA UNIVERSITY : CHENNAI 600 025

MARCH 2026

KARPAGAM
COLLEGE OF ENGINEERING
Rediscover | Refine | Redefine
(Autonomous)

BONAFIDE CERTIFICATE

Register Number: _____

Certified that this is the Bonafide Record of work done by Mr./Ms._____ of III semester B.E Electrical and Electronics Engineering (Branch) during the academic year _____ in the 23CDO104 – System Programming in C Laboratory.

Date: **Faculty in-charge**

Register Number: _____

Submitted for the End Semester Practical Examination held on _____at **Karpagam College of Engineering, Coimbatore**.

SIGNATURE
Dr.K.SHEELA SOBANA RANI
Professor,
Domain Head - Firmware,
Karpagam Institutions

Coimbatore. Tamilnadu.

SIGNATURE
Dr. MOHAN KUMAR J
Professor,
Domain Incharge - Firmware,
Karpagam Institutions,

Coimbatore. Tamilnadu.

KARPAGAM
COLLEGE OF ENGINEERING
Rediscover | Refine | Redefine
(Autonomous)

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

VISION

To provide students with sound knowledge of Electrical and Electronics Engineering, that they become capable of facing the current and impending challenges to extend their expertise in the global arena.

MISSION

- Impart high quality education and training to the students in the field of Electrical and Electronics Engineering.
- Transforming our students into enterprising technologists by giving them excellent facilities by qualified, committed members. of faculty.
- Motivating them to contribute immensely for the benefit of the entire humanity

Programme Educational Objectives (PEOs)

1. **PEO1:** Graduates will synthesize mathematics, science, engineering fundamentals, laboratory and work-based experiences to formulate and solve problems in Electrical and Electronics engineering and the related domains and will develop proficiency in Computer-based engineering and the use of computational tools.
2. **PEO2:** Graduates will communicate and work team-based on the multidisciplinary engineering projects in the allied fields of Electrical Science and will practice the ethics of their profession.
3. **PEO3:** Graduates will realize the importance of self-learning and engage in lifelong learning to become experts either as an entrepreneur or an employee so as to broaden their knowledge in the domain.

Program Outcomes

- **PO1: Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **PO2: Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

- **PO3: Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- **PO4: Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- **PO5: Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- **PO6: The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- **PO7: Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **PO8: Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **PO9: Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **PO10: Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- **PO11: Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- **PO12: Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

After successful completion of the program, graduates of B.E (EC) will:

PSO-1 Apply the knowledge in the field of electrical and electronics engineering to analyse, design and develop solutions for real world problems.

PSO-2 Demonstrate the skill in core and allied domain to work in interdisciplinary teams.

PART A – DS

S.NO	LIST OF EXPERIMENTS
1	Write a C program to create a playlist in an audio player using linked list.
2	Write a C program to create a Congo line using linked list.
3	Write a C program to implement scavenger hunt or treasure hunt using linked list.
4	Write a C program to implement the elimination game (Josephus problem) using linked list where every kth player is removed until one player remains.
5	Write a C program to implement the history list of file directories and traverse to previous or next directory using linked list.
6	Write a C program to implement hash tables using linked list (chaining method).
7	Write a C program to implement a tree to identify the car manufacturer and car type by providing car name.
8	Write a C program to implement a Graph using linked list (Adjacency List representation).
9	Write a C program to implement a Scheduler using linked list.
10	Write a C program to implement string reversal using stack.
11	Write a C program to check balanced parentheses using stack.
12	Write a C program to implement undo operation using stack.
13	Write a C program to implement token number announcement system model for banks and hospitals using queue.
14	Write a C program to implement reservation or ticket counter model using queue.
15	Write a C program to implement priority-based scheduling using heap sort.

PART B -OS

S.NO	LIST OF EXPERIMENTS
1	Create two processes: one reads a thermostat sensor, the other controls a compressor motor. Use pipes for communication.
2	Implement three processes: a state machine sends commands, one controls a motor, one controls a valve. Use pipes for command transfer.
3	Develop three processes: image writer (JPG), video writer (3GP), and display process.
4	Create an application to read records via serial communication after handshaking, write them to a file, and sort the table.
5	Build three processes: angle sensor reader, proximity sensor reader, and motor controller via serial port. Use message queues.
6	Design four processes: heater controller, light/door monitor, fan/turntable controller, and thermostat reader. Use message queues.
7	Create four processes: time reader, temperature reader, humidity reader, and display process. Use shared memory.
8	Implement three processes: pressure sensor reader, pump controller, and calculator/display. Use message queues + signals.
9	Build three processes: user input reader (rate/time), calculator controlling pump, and pressure sensor reader. Use message queues.
10	Implement three processes: key input handler, audio player, and photo frame. Use message queues, interrupts, signals.
11	Create three processes: user input handler, sequential motor controller, and periodic calculator. Use semaphores, message queues, signals, shared memory.
12	Implement multiple threads: runtime updater, charging tracker, battery usage tracker, sensor1 logger, sensor2 logger, event log writer, and display thread. Use shared memory + mutex.
13	Build an application to read a configuration file and execute device cycles (pump, fluid, door, drain).
14	Create a file manager application in Linux to perform file operations (create, delete, copy, move, search).
15	Implement networking application to control home appliances: one process receives server commands, another sends control signals to devices.

PART C - BAREMETAL

S.NO	LIST OF EXPERIMENTS
1	Write an ARM Cortex-M assembly program to add two 32-bit numbers and store the result in memory.
2	Implement an ARM Cortex-M assembly program to generate Fibonacci series up to 10 terms and store them in RAM.
3	Develop an ARM Cortex-M assembly program to toggle a GPIO pin using bit manipulation instructions.
4	Write an ARM Cortex-M assembly program to implement a delay loop using registers and branch instructions.
5	Write an ARM Cortex-M assembly program to implement a delay loop using registers and branch instructions.
6	Interface an LED with ARM Cortex board and write a C program to blink using GPIO.
7	Interface a 16×2 LCD and display “HELLO WORLD” using GPIO in 4-bit mode
8	Control a DC motor using ARM Cortex board and change direction (forward/reverse) via GPIO.
9	Write a C program to implement a Scheduler using linked list.
10	Write a C program to implement string reversal using stack.
11	Write a C program to check balanced parentheses using stack.
12	Write a C program to implement undo operation using stack.
13	Write a C program to implement token number announcement system model for banks and hospitals using queue.
14	Write a C program to implement reservation or ticket counter model using queue.
15	Write a C program to implement priority-based scheduling using heap sort

PART A – DS

Exp.No	Preparation (40)	Execution (40)	Viva Voce (20)	Total (100)	Signature
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					

PART B – OS

Exp.No	Preparation (40)	Execution (40)	Viva Voce (20)	Total (100)	Signature
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					

PART C – BAREMETAL

Exp.No	Preparation (40)	Execution (40)	Viva Voce (20)	Total (100)	Signature
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					